

Total No. of Questions : 4]

SEAT No. :

[Total No. of Pages : 1

P8467

Oct-22/BE/Insem-43

B.E. (Computer Engineering)

BLOCKCHAIN TECHNOLOGY

(2019 Pattern) (Semester-VII) (410243)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) List and explain advantages of ECC. [5]
- b) Differentiate asymmetric and symmetric key cryptography. [5]
- c) Discuss the properties of hash function. [5]

OR

- Q2)** a) List and discuss benefits of Merkel tree. [5]
- b) Explain DSA key generation and verification. [5]
- c) Discuss role of hashing in Blockchain. [5]

- Q3)** a) Explain consensus layer in Blockchain. [5]
- b) Discuss limitations of centralised systems. [5]
- c) Why Blockchain is important? [5]

OR

- Q4)** a) Explain propagation layer in blockchain. [5]
- b) Discuss evolution of Blockchain. [5]
- c) Differentiate centralised & decentralised systems. [5]



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SEAT No. :

P-5191

[Total No. of Pages : 1

[6188]-143

B.E. (Computer Engineering) (Insem)
BLOCK CHAIN TECHNOLOGY
(2019 Pattern) (Semester - VII) (410243)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Solve Q1 or Q2, Q3 or Q4.*
- 2) *Figures to the right indicate full marks.*
- 3) *Assume Suitable data, if necessary.*
- 4) *Neat diagrams must be drawn wherever necessary.*

- Q1)** a) Illustrate Elliptic Curve Cryptography. [6]
b) Justify the importance of Hashing in Block Chain. [4]
c) What is Merkle tree? Explain the structure of merkle tree. [5]

OR

- Q2)** a) Explain working of SHA 256 Algorithm. [6]
b) Describe Digital Signature & Verification steps in Digital Signature Algorithm. [4]
c) Describe Symmetric Key Encryption with neat diagram. [5]

- Q3)** a) Discuss various limitations of centralized system with respect to Decentralized system. [6]
b) Write a note on Evolution of Block Chain. [4]
c) List and Explain algorithms of Consensus layer. [5]

OR

- Q4)** a) Write a note on : [6]
i) Propagation layer
ii) Application layer
b) List & Explain features of Block Chain. [4]
c) Comment on "Feasibility of an Online Voting System Implementation" using Block Chain Technology. [5]



Total No. of Questions : 4]

SEAT No. :

PC185

[6361]-43

[Total No. of Pages : 1

**B.E. (Computer Engineering) (Insem)
BLOCKCHAIN TECHNOLOGY
(2019 Pattern) (Semester-VII) (410243)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary.*

- Q1)** a) Differentiate between symmetric and asymmetric key cryptography. [5]
- b) What is hashing? Explain role of hashing in Blockchain. [5]
- c) What is Merkle tree? Explain with diagram. [5]

OR

- Q2)** a) Explain digital signature algorithm. [5]
- b) What are the benefits of Merkle tree in Blockchain. [5]
- c) Discuss elliptic curve cryptography. [5]

- Q3)** a) Explain consensus Layer in Blockchain. [5]
- b) Differentiate between centralized and decentralized systems. [5]
- c) Discuss evolution of Blockchain. [5]

OR

- Q4)** a) Explain semantic Layer in Blockchain. [5]
- b) Discuss importance of Blockchain. [5]
- c) What is decentralized system? Explain with diagram. [5]



Total No. of Questions : 4]

SEAT No. :

PE-187

[Total No. of Pages : 2

[6580]-547

B.E. (Computer Engineering) (Insem.)
BLOCK CHAIN TECHNOLOGY
(2019 Pattern) (Semester - VII) (410243)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figure to the right indicates full marks.*

- Q1)** a) Explain the working of SHA-256 Algorithm. [6]
b) Describe asymmetric key encryption with neat diagram. [4]
c) What is Merkle tree? Explain the structure of merkle tree. [5]

OR

- Q2)** a) Illustrate Elletptic curve cryptography with detail steps. [6]
b) How digital signature & Verification is carried out in digital signature algorithm. [4]
c) List and explain the features of hashing functions. [5]

- Q3)** a) Define the terms with suitable example. [6]
i) Consensus
ii) Distributed ledger
b) List & explain features of Block chain. [4]
c) Write a note on Propagation Layer & Application Layer. [5]

OR

P.T.O.

- Q4)** a) Discuss various limitations of centralized system with respect to De-centralized system. [6]
- b) Explain the Evolution of Block Chain with time line. [4]
- c) What do you think, which limitations of block chain are major hurdle's in its adoption. [5]

